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(54) **COSMETIC PRODUCT APPLICATOR, COSMETIC PRODUCT PACKAGING ASSEMBLY AND ASSOCIATED MANUFACTURING PROCESSES**

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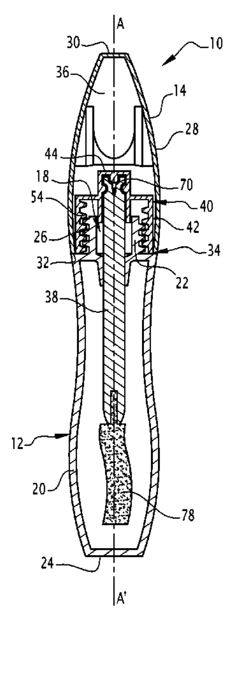
(57) **ABSTRACT**

A cosmetic product application member comprises:
a rod carrier (**40**); and
a rod (**38**) defining a first end (**70**) and a second end (**72**) that are opposite.
(Continued)



The first end (**70**) of the rod (**38**) is intended to be assembled in the rod carrier (**40**). The rod (**38**) and the rod carrier (**40**) are manufactured from a plastic material composed of a copolyester, in particular a semi-aromatic copolyester.

17 Claims, 5 Drawing Sheets

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B29K 67/00	(2006.01)
B29L 31/00	(2006.01)

(52) **U.S. Cl.**
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See application file for complete search history.

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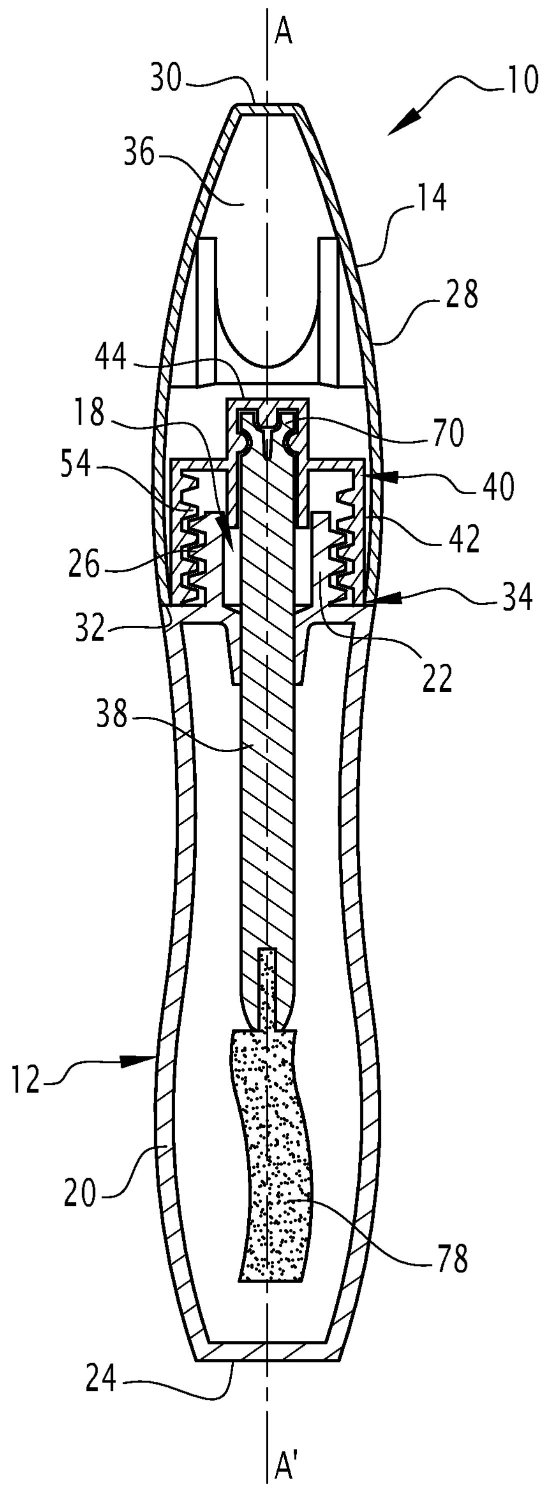

FIG.1

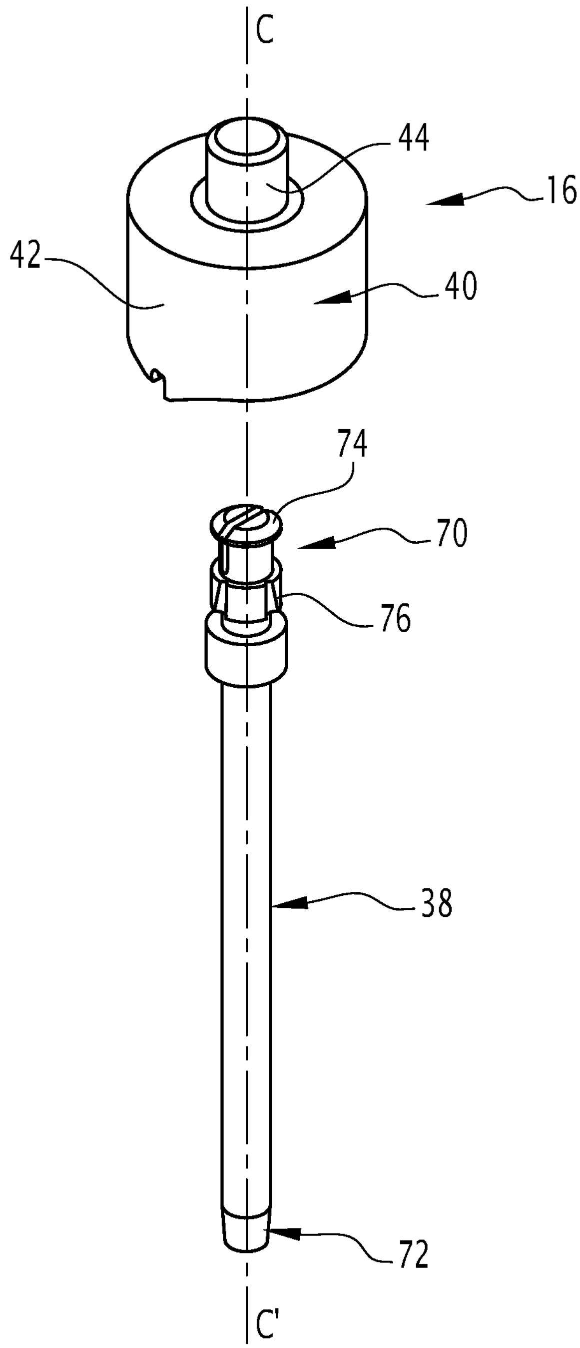

FIG.2

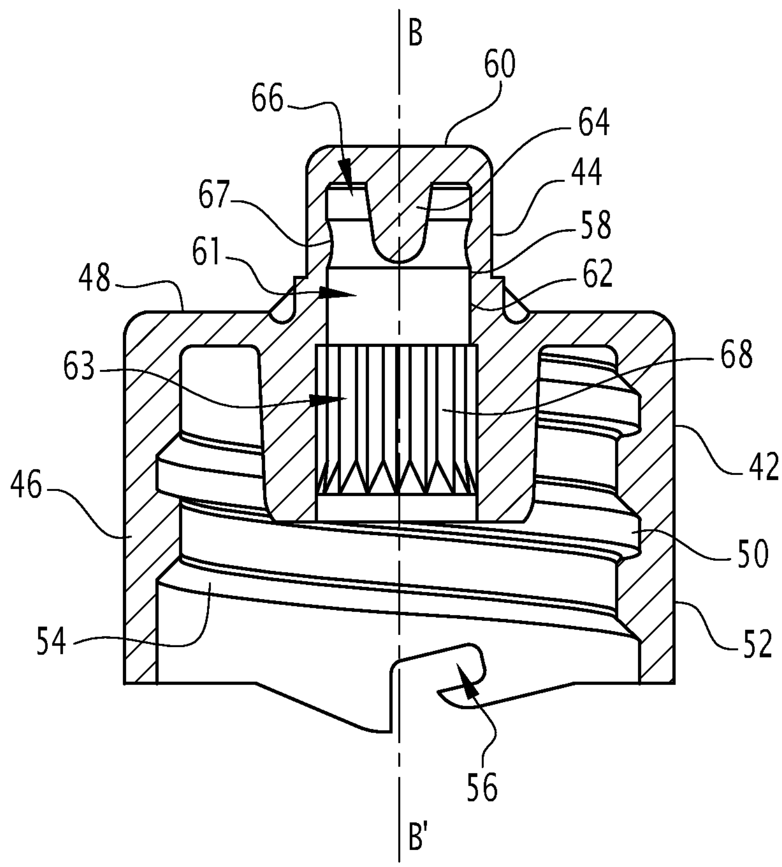

FIG.3

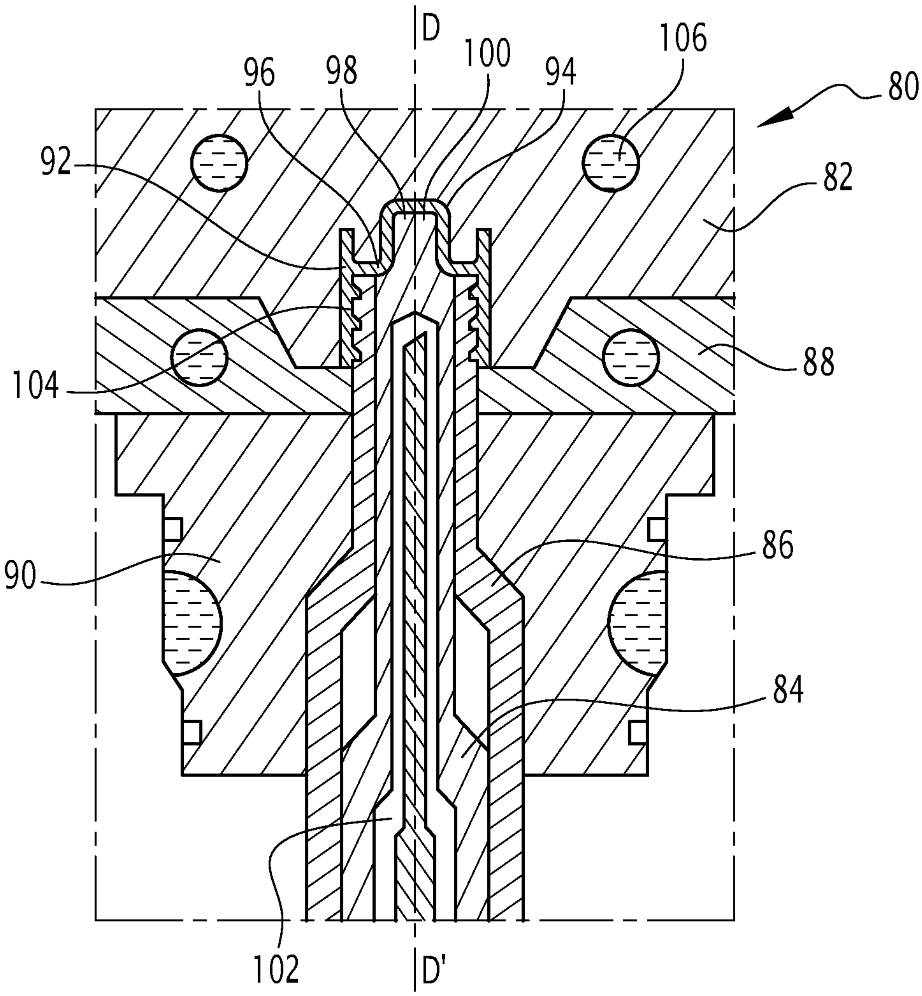

FIG.4

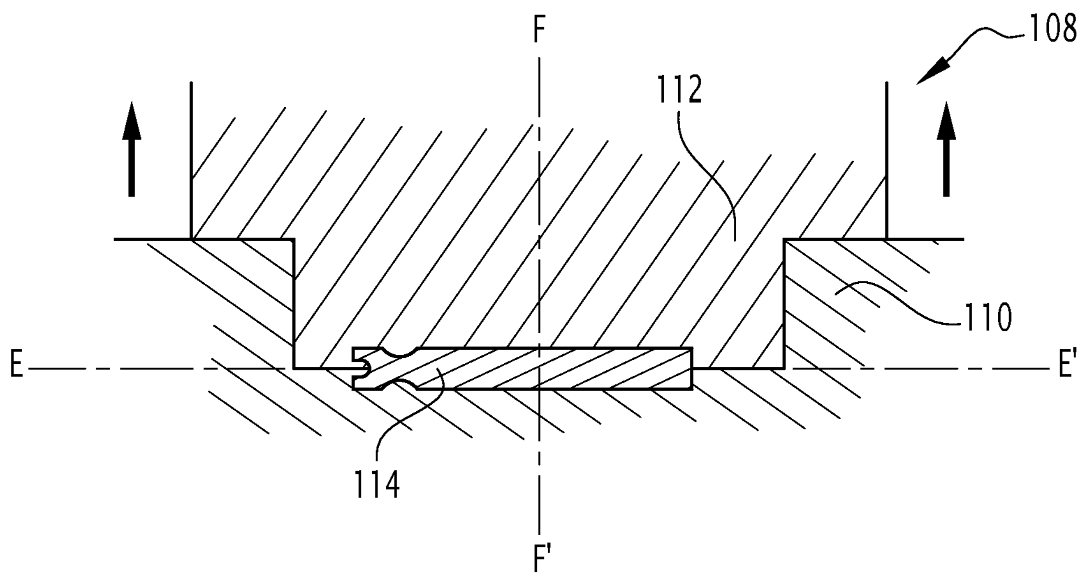

FIG.5

COSMETIC PRODUCT APPLICATOR, COSMETIC PRODUCT PACKAGING ASSEMBLY AND ASSOCIATED MANUFACTURING PROCESSES

CROSS REFERENCE TO RELATED APPLICATIONS

This application is a National Phase filing under 35 U.S.C. § 371 of PCT/EP2022/065689, filed on Jun. 9, 2022, which in turn claims priority to French Application No. 2106132, filed Jun. 10, 2021, disclosures of which are incorporated herein by reference in their entireties.

The present invention relates to a cosmetic product application member comprising a rod carrier and a rod defining a first end and a second end that are opposite.

A cosmetic product is in particular a product as defined in EC Regulation N° 1223/2009 of the European Parliament and the Council of Nov. 30, 2009, relating to cosmetic products. For example, the cosmetic product is a coloring, make-up or care product, such as lip gloss or mascara.

The cosmetic product is generally contained in a receptacle or a flask equipped with a cosmetic product application member capable of extracting a quantity of cosmetic product and applying said quantity of product on a body surface, so as to obtain the desired effect.

The application member generally comprises a rod intended to carry an applicator intended to reach the base of the receptacle and a rod carrier intended to mount the rod on the receptacle.

It is known to manufacture such application members in one piece molded in a plastic material, such as for example polyoxymethylene (POM).

However, the plastic materials which are simple and inexpensive to mold such as POM have no recycling processes allowing reuse of the application member which protects the environment.

Plastic materials composed of a copolyester, such as polyethylene terephthalate (PET) for example, offer the advantage of being easy to recycle and having well-established industrial recycling processes.

However, the manufacture of an application member by molding a copolyester is costly in cycle time, due to the substantial cooling time of such a material. For want of such cooling, the released product is relatively soft, which causes problems in terms of the mechanical strength thereof and meeting the desired dimensional specifications.

Such a manufacture is therefore expensive and unsuitable for mass production.

The problems caused by the substantial cooling time of copolyesters are particularly important in the context of molding an elongated element, such as a rod, released by moving a mold along the longitudinal axis thereof; indeed, in the absence of complete cooling of the elongated element, the latter will tend to adhere to the walls of the mold and elongate during such a mold release operation, resulting in a failure to comply with the manufacturing dimensions sought, or even impossible mold release.

An aim of the invention is that of providing a cosmetic product application member made of an easy-to-recycle material, while remaining simple and inexpensive to manufacture, and meeting the desired dimensional specifications.

For this purpose, the invention relates to a cosmetic product application member of the type cited above, characterized in that the first end of the rod is intended to be assembled in the rod carrier, the rod and the rod carrier being manufactured from a plastic material composed of a copolyester, in particular a semi-aromatic copolyester.

Such an application member is manufactured from an easy-to-recycle material that has well-established industrial recycling processes. Manufacturing such an application member in two parts offers the advantage of remedying the problem of the substantial cooling time and therefore of expensive manufacture. Indeed, by designing the application member in two parts, the rod and the rod carrier can be manufactured, and in particular molded, separately, in molding devices adapted to the geometry thereof, particularly enabling optimal cooling at least cost in terms of cycle time of the molded copolyester material. Such an application member is thus simple and inexpensive to manufacture, while ensuring that it meets the desired dimensional specifications.

In a specific embodiment of the invention, the plastic material is composed of virgin polyethylene terephthalate (PET), recycled PET or a combination of virgin PET and recycled PET, and preferably more than 90% recycled PET.

Such a material has good mechanical properties and offers the advantage of having a very well-established industrial recycling process. The application member can thus be manufactured solely from recycled materials.

In a specific embodiment of the invention, the first end of the rod is intended to be snap-locked into the rod carrier.

Snap-locking the rod in the rod carrier enables reliable fastening of the rod in the rod carrier, while remaining simple to carry out. The embodiment of the application member in two parts, making it possible to obtain a rod carrier and a rod of precise geometries, is especially advantageous in the context of a rod assembled by snap-locking on the rod carrier, the reliability of such an assembly device being conditional on complying with precise manufacturing tolerances for these two elements.

In a specific embodiment of the invention, the rod carrier comprises:

- a screwing ring centered on a rod carrier axis and defining an internal thread; and
- a cylindrical skirt centered on the rod carrier axis, extending at least partially inside the screwing ring and intended to receive the first end of the rod.

In a specific embodiment of the invention, the cylindrical skirt comprises a protuberance centered on the rod carrier axis, protruding inside the cylindrical skirt and defining at least one retaining housing intended to receive at least one tenon arranged at the first end of the rod.

In a specific embodiment of the invention, the cylindrical skirt comprises longitudinal ribs arranged along at least a portion of an inner surface of the cylindrical skirt, the rod comprising at least one longitudinal protrusion capable of engaging between two adjacent longitudinal ribs of the cylindrical skirt.

Such a structure of the rod carrier and rod is adapted for snap-locking the rod in the rod carrier, while enabling fastening to a cosmetic product packaging assembly.

The invention also relates to a cosmetic product packaging assembly comprising:

- a receptacle intended to receive the cosmetic product and defining an opening;
- a cover intended to close the opening; and
- a cosmetic product application member as defined above, the rod carrier being intended to be fastened in the cover.

In a specific embodiment of the invention, the receptacle, the cover, the rod carrier and the rod are manufactured from the same material.

Manufacturing the elements of the cosmetic product packaging assembly from the same material enables simplified recycling thereof.

The invention also relates to a process for manufacturing a rod carrier, the rod carrier being intended to be assembled with a rod to form a cosmetic product application member, the manufacturing process including the following steps:

- providing a rod carrier molding device comprising a die and a central spindle and defining a rod carrier molding cavity; and
- injecting a fluid material into the molding cavity;
- solidifying the fluid material in the molding cavity to form the rod carrier.

The fluid material injected into the molding cavity is a copolyester, particularly a semi-aromatic copolyester.

In a specific embodiment of the invention, during the solidification step, the central spindle is cooled by injecting a coolant into a channel defined inside the central spindle.

Such cooling of the central spindle makes it possible to cool the fluid material injected into the molding cavity effectively and thus reduce the solidification time thereof. The rod carrier thus obtained is produced at a low cost in terms of cycle time, while ensuring that the desired dimensional specifications are met.

In a specific embodiment of the invention, the molding device further comprises a threaded spindle; during the solidification step, the threaded spindle being cooled by thermal contact with the central spindle.

The injected fluid material is composed of virgin polyethylene terephthalate, recycled PET or a combination of virgin PET and recycled PET, and preferably more than 90% recycled PET.

The invention also relates to a process for manufacturing a cosmetic product application member including the following steps:

- manufacturing a rod carrier according to the manufacturing process described above;
- providing a rod defining a first end and a second end that are opposite, the rod being composed of the same material as the rod carrier; and
- assembling the first end of the rod in the rod carrier.

In a specific embodiment of the invention, the first end of the rod is snap-locked into the rod carrier.

The rod is molded flat beforehand.

Such flat molding of the rod, which is presented in the form of an elongated element, is particularly advantageous in terms of investment, the mold used being of simple design, for example in two shells, and in terms of cooling time required for mold release without adhesion of the molded copolyester material to the walls of the mold and meeting the desired manufacturing dimensions.

Indeed, in the case of flat molding of an elongated element, mold release is performed by moving the mold parts in an orthogonal direction to the longitudinal axis of the molded elongated element and not along this longitudinal axis. This prevents the pitfall of stretching the rod along the longitudinal axis thereof during mold release, due to the phenomenon of adhesion of the material to the walls of a mold moving along this longitudinal axis. Moreover, a device for flat molding such an elongated element is relatively easy to cool, which further reduces the cooling time needed to obtain a rod of precise geometry after mold release.

The invention will be easier to understand after reading the following description, provided solely as an example, and with reference to the appended drawings, wherein:

FIG. **1** is a cross-section along an axial midplane of a cosmetic product packaging assembly comprising a cosmetic product application member according to the invention;

FIG. **2** is an exploded view of the cosmetic product application member in FIG. **1** comprising a rod carrier and a rod;

FIG. **3** is a cross-section along an axial midplane of the rod carrier in FIG. **2**;

FIG. **4** is a schematic cross-section, along an axial midplane of the rod carrier molding device in FIG. **2**; and

FIG. **5** is a schematic cross-section, along an axial midplane of the rod molding device in FIG. **2**.

FIG. **1** illustrates a cosmetic product packaging assembly **10** comprising a receptacle **12**, a cover **14** and a cosmetic product application member **16**.

The cosmetic product intended to be packaged with the assembly **10** is for example, a cosmetic care, coloring or make-up product. It is presented in liquid or pasty form. The cosmetic product is for example a mascara, a lip gloss, or a nail varnish.

The receptacle **12** is intended to receive said cosmetic product and defines an opening **18**.

The receptacle **12** is for example a flask or a bottle.

The receptacle **12** extends longitudinally along a main axis A-A'. Under normal conditions of use, the main axis A-A' corresponds to a top-bottom axis.

The receptacle **12** comprises a side wall **20**, a neck **22** and a base **24** located opposite the neck **22**.

The side wall **20** is a hollow tube of main axis A-A', as illustrated in FIG. **1**.

The volume defined inside the side wall **20**, delimited at the top by the neck **22** and the bottom by the base **24** is capable of receiving the cosmetic product.

The internal volume of the receptacle **12** has for example a capacity between 4 mL and 20 mL.

As illustrated in FIG. **1**, the neck **22** defines the opening **18**, which is advantageously centered on the main axis A-A', and which opens at the top.

The opening **18** is for example of circular shape of diameter between 3 mm and 15 mm.

The neck **22** is advantageously cylindrical of circular cross-section centered on the main axis A-A'.

An external thread **26** is advantageously arranged on the neck **22** as illustrated in FIG. **1**.

The receptacle **12** is for example made of plastic manufactured by plastic injection or blowing.

Preferably, the receptacle **12** is manufactured from a plastic material composed of a copolyester, particularly a semi-aromatic copolyester and more particularly from a plastic material composed of virgin polyethylene terephthalate (PET), recycled PET or a combination of virgin PET and recycled PET, and preferably more than 90% recycled PET.

The cover **14** is intended to close the opening **18** defined in the neck **22** of the receptacle **12**.

The cover **14** extends longitudinally along a main axis, which is merged with the main axis A-A', when the cover **14** closes the opening **18** as illustrated in FIG. **1**.

The cover **14** comprises a side wall **28**, a closing wall **30** located at a first end of the cover **14** located away from the receptacle **12**, and an annular shoulder **32** located at a second end of the cover **14** and defining an opening **34**.

The cover **14** is advantageously hollow and defines an inner volume **36** inside the side wall **28**, delimited at the top by the closing wall **30** and open at the bottom via the opening **34**.

When the cover **14** closes the opening **18**, the shoulder **32** is in contact with the receptacle **12** and the neck **22** is located in the inner volume **36** of the cover **14**.

The opening **34** defined by the shoulder **32** is for example of circular shape of diameter between 4 mm and 15 mm.

The cover **14** is for example manufactured by plastic injection or blowing.

Advantageously, the cover **14** is manufactured from the same material as the receptacle **12**, and preferably from a plastic material composed of a copolyester, particularly a semi-aromatic copolyester and more particularly from a plastic material composed of virgin PET, recycled PET or a combination of virgin PET and recycled PET, and preferably more than 90% recycled PET.

The cosmetic product application member **16** comprises a rod **38** intended to be inserted into the receptacle **12** and a rod carrier **40** intended to carry the rod **38**, while being assembled with the cover **14**.

As seen hereinafter, the rod **38** and the rod carrier **40** are made of two separate parts assembled with one another. They are each made of a plastic material composed of a copolyester, particularly a semi-aromatic copolyester and more particularly of a plastic material composed of virgin PET, recycled PET or a combination of virgin PET and recycled PET, and preferably more than 90% recycled PET.

The cosmetic product application member **16** is movably mounted through the opening **18** along an insertion direction between an extraction position and an application position.

The insertion direction extends for example along the main axis A-A' of the receptacle **12**.

In the extraction position, as illustrated in FIG. **1**, the cosmetic product application member **16** is located inside the receptacle **12** so as to be at least partially in contact with the cosmetic product.

In the sampling position, the cover **14** closes the opening **18** and the rod **38** is located inside the receptacle **12**.

In the application position, the cover **14** leaves the opening **18** of the receptacle **12** free and open, the cosmetic product application member **16** is extracted from the receptacle **12** and is capable of being moved in the vicinity of or in contact with a body surface, with a view to applying an extracted quantity of cosmetic product thereon.

The rod carrier **40** is fastened in the cover **14**, such that the application member **16** is integral with the cover **14**.

As illustrated in FIG. **1**, the rod carrier **40** is inserted into the inner volume **36** via the opening **34** of the cover **14**.

Once fastened, the rod carrier **40** is disposed bearing laterally on the inner surface of the side wall **28** of the cover **14**.

In the example illustrated in FIGS. **2** and **3**, the rod carrier **40** comprises a screwing ring **42** and a cylindrical upper skirt **44** inserted through the screwing ring **42**.

The screwing ring **42** is centered on a rod carrier axis B-B' and comprises a side wall **46** and an upper wall **48**.

The side wall **46** is a hollow sleeve centered on the rod carrier axis B-B'. It defines the inner surface **50** and an outer surface **52** which are substantially cylindrical.

The screwing ring **42** is intended to be screwed onto the neck **22** of the receptacle **12**, such that the cover **14**, wherein the rod carrier **40** is mounted, closes the opening **18**.

For this purpose, the screwing ring **42** defines an internal thread **54** capable of being screwed onto the external thread **26** of the neck **22** of the receptacle **12**.

As illustrated in FIG. **3**, the internal thread **54** is defined on the inner surface **50** of the side wall **46**.

The outer surface **52** of the side wall **46** of the screwing ring **42** is intended to bear laterally on the inner surface of the side wall **28** of the cover **14**, so as to fasten the rod carrier **40** into the cover **14**.

The screwing ring **42** has for example an inner diameter substantially equal to the diameter of the opening **18** of the receptacle **12** and an outer diameter substantially equal to the diameter of the opening **34** of the cover **14**.

When the screwing ring **42** is screwed onto the neck **22** of the receptacle **12**, the rod carrier axis B-B' is substantially merged with the main axis A-A'.

Advantageously, as illustrated in FIG. **3**, a hook **56** is arranged on the screwing ring **42**. The hook **56** is capable of snap-locking onto a stop defined for example on the neck **22** of the receptacle **12**. At the end of the screwing travel of the screwing ring **42** on the neck **22**, such snap-locking makes it possible to offer the user an audible and tactile closing sensation.

The cylindrical skirt **44** is centered on the rod carrier axis B-B' and comprises a side partition **58** and an upper partition **60**.

The side partition **58** is a hollow tube centered on the rod carrier axis B-B' and defines a substantially cylindrical inner surface **62**.

The cylindrical skirt **44** extends at least partially inside the screwing ring **42**.

More particularly, as illustrated in FIG. **3**, the upper partition **60** of the cylindrical skirt **44** extends substantially parallel with the upper wall **48** of the screwing ring **42** and away therefrom.

The cylindrical skirt **44** comprises an upper part **61**, wherein the side partition **58** protrudes upwards to the upper partition **60** of the cylindrical skirt **44** beyond the upper wall **48** of the screwing ring **42**. It connects the cylindrical skirt **44** and the screwing ring **42**.

The cylindrical skirt **44** also comprises a lower part **63**, wherein the side partition **58** extends protruding downwards inside the screwing ring **42**.

The cylindrical skirt **44** has for example an outer diameter less than the diameter of the screwing ring **42**, and particularly between 3 mm and 10 mm.

As illustrated in FIG. **3**, the cylindrical skirt **44** further comprises a protuberance **64** centered on the rod carrier axis B-B', protruding toward the inside of the cylindrical skirt **44** and defining at least one retaining housing **66**.

More particularly, the protuberance **64** is arranged under the upper partition **60** and has a rounded tip centered on the rod carrier axis B-B'.

The cylindrical skirt **44** also advantageously comprises an annular retaining bead **67** protruding from the side partition **58** toward the inside of the cylindrical skirt **44**.

As illustrated in FIG. **3**, the retaining housing **66** has an annular shape. It is delimited laterally between the protuberance **64** and the side partition **58** and is delimited along the rod carrier axis B-B' between the annular retaining bead **67** and the upper partition **60** of the cylindrical skirt **44**. The retaining housing **66** is for example defined at the upper part **61** of the cylindrical skirt **44**.

The cylindrical skirt **44** further comprises longitudinal ribs **68** arranged on at least a portion of the inner surface **62** of the cylindrical skirt **44**.

The longitudinal ribs **68** extend along a parallel direction with the rod carrier axis B-B', and for example as illustrated in FIG. **3**, only on the lower part **63** of the cylindrical skirt **44**.

The rod carrier **40** is for example manufactured by plastic injection molding.

Preferably, the rod carrier **40** is manufactured from a plastic material composed of a copolyester, particularly a semi-aromatic copolyester and more particularly from a plastic material composed of virgin PET, recycled PET or a combination of virgin PET and recycled PET, and preferably more than 90% recycled PET.

Advantageously, the rod carrier **40** is manufactured from the same material as the receptacle **12** and the cover **14**.

As illustrated in FIG. **2**, the rod **38** extends in a rectilinear manner along a rod axis C-C'.

The rod **38** defines, along the rod axis C-C', a first end **70** and a second end **72** opposite the first end **70**.

The first end **70** of the rod **38** is intended to be assembled in the rod carrier **40**, as illustrated in FIG. **1**.

More particularly, the first end **70** of the rod **38** is intended to be snap-locked in the rod carrier **40**.

In other words, the rod carrier **40**, and more particularly the cylindrical skirt **44** of the rod carrier **40**, is capable of receiving the first end **70** of the rod **38** and locking it in position.

For this purpose, the retaining housing **66** defined in the cylindrical skirt **44** of the rod carrier **40** is intended to receive two tenons **74** arranged at the first end **70** of the rod **38**.

The two tenons **74** extend along a parallel direction with the rod axis C-C' and give the first end **70** of the rod **38** a fork shape.

More particularly, during the insertion of the first end **70** of the rod **38** in the cylindrical skirt **44**, the tenons **74** are capable of being deformed elastically toward the rod axis C-C', so as to be housed in the retaining housing **66**, then return to an idle position, wherein each tenon **74** is locked thus in the retaining housing **66**. Such shape cooperation locks the first end **70** of the rod **38** in position in the rod carrier **40**.

When the rod **38** is assembled in the rod carrier **40**, the rod carrier axis B-B' and the rod axis C-C' are substantially merged.

Advantageously, the rod **38** further comprises longitudinal protrusions **76** arranged on the first end **70** of the rod **38**, below the tenons **74**.

The longitudinal protrusions **76** extend along a parallel direction with the rod axis C-C.

The longitudinal protrusions **76** are capable of being housed between longitudinal ribs **68** of the rod carrier **40** when the first end **70** of the rod **38** is assembled in the rod carrier **40**.

Such an assembly of the protrusions **76** in the ribs **68** locks the rotation of the rod **38** in the rod carrier **40**, which could damage the snap-locking connection between the rod **38** and the rod carrier **40**, particularly when screwing and unscrewing the screwing ring **42** on the neck **22** of the receptacle **12**.

Alternatively, the assembly of the first end **70** of the rod **38** in the rod carrier **40** is performed by welding, particularly by ultrasonic welding.

The rod **38** is for example manufactured from a plastic material composed of a copolyester, particularly a semi-aromatic copolyester and more particularly from a plastic material composed of virgin PET, recycled PET or a combination of virgin PET and recycled PET, and preferably more than 90% recycled PET.

Advantageously, the rod **38** is manufactured from the same material as the rod carrier **40**, and preferably as the receptacle **12** and the cover **14**.

The cosmetic product application member **16** further comprises an applicator **78** fastened at the second end **72** of the rod **38**.

The applicator **78** is capable of extracting a quantity of cosmetic product and applying said quantity of product on a body surface so as to obtain the desired effect.

The applicator **78** is adapted to the cosmetic product to be applied.

The applicator **78** is for example a mascara brush as illustrated in FIG. **1**, or is for example a flocked brush intended to apply lip gloss.

A rod carrier molding device **80** intended to mold the rod carrier **40** will now be described with reference to FIG. **4**.

The molding device **80** comprises a die **82**, a central spindle **84**, a peripheral threaded spindle **86**, two deux carriages **88** and a steady **90**. It defines a molding cavity **92** of the rod carrier **40**.

The molding cavity **92** has a shape and size conjugated with the rod carrier **40**.

The molding cavity **92** comprises an upper face **94** defining the outer contour of the rod carrier **40** and a lower face **96** defining the inner contour of the rod carrier **40**.

The molding cavity **92** is centered on a central axis D-D'. Under normal conditions of use, the central axis D-D' corresponds to a top-bottom axis of the molding device **80**.

The molding cavity **92** is capable of receiving a fluid plastic material injected using a nozzle (not illustrated) and which, on solidifying, forms a rod carrier **40**.

The die **82** is movable in translation along the central axis D-D'.

The upper face **94** of the molding cavity **92** is arranged in said die **82**.

The central spindle **84** extends along the central axis D-D' and comprises an upper end **98** defining at least partially the lower face **96** of the molding cavity **92**.

The central spindle **84** is movable in translation without rotation along the central axis D-D' and to define the molding cavity **92**, the central spindle **84** is capable of being inserted along the central axis D-D' in the direction of the die **82**.

Optionally, an end piece **100** is fastened at the upper end **98** of the central spindle **84**. Such an end piece **100** makes it possible for example to define the complex shape of the cylindrical skirt **42** of the rod carrier **40**.

According to the invention, a channel **102** is defined inside the central spindle **84**.

The channel **102** is fluidically connected to a coolant source, in particular to a source of water at a temperature less than 40° C.

The channel **102** extends along the central axis D-D' toward the upper end **98** of the central spindle **84**.

The threaded spindle **86** has a hollow cylindrical shape centered on the central axis D-D' and comprises an upper end **104** capable of defining the internal thread **54** of the screwing ring **42** of the rod carrier **40**.

As illustrated in FIG. **4**, the central spindle **84** is disposed inside the threaded spindle **86**, so as to define the lower face **96** of the molding cavity **92** together.

The central spindle **84** and the threaded spindle **86** are thus in thermal contact with one another, in particular to enable thermal conduction. Thus, cooling of the central spindle **84** by circulating a coolant in the channel **102** induces cooling of the threaded spindle **86**.

The carriages **88** and the steady **90** are disposed around the spindles **84**, **86**, and in particular the threaded spindle **86**, so as to hold the spindles **84**, **86** in position during molding.

The carriages **88** are movable in translation along a perpendicular axis to the central axis D-D' and the steady **90** is movable in translation along the central axis D-D'.

The fluid material present in the molding cavity **92** is capable of being cooled by thermal contact with the central spindle **84** and the threaded spindle **86** cooled by circulating a coolant in the channel **102** defined in the central spindle **84**.

Moreover, cooling circuits **106** are defined in the die **82**, the carriages **88** and steady **90** so as to cool the fluid material injected into the molding cavity **92** further.

A process for manufacturing a cosmetic product application member **16** according to the invention will now be described.

A rod carrier **40** is first manufactured according to the following process for manufacturing the rod carrier.

Firstly, a rod carrier molding device **80** as described above is provided.

The different components of the molding device **80** are arranged so as to define the molding cavity **92**.

For this purpose, the die **82** is fastened in position. The spindles **84**, **86** are then translated along the central axis D-D' in the direction of the die **82** so as to define the molding cavity **92**. The carriages **88** and the steady **90** are disposed so as to hold the spindles **84**, **86** in position.

Secondly, a fluid material intended to solidify is injected into the molding cavity **92**, for example via a nozzle (not illustrated).

The injected fluid material is a thermoplastic material and more particularly a plastic material composed of a copolyester, which has been previously heated for example to a temperature greater than 240° C.

Preferably, the injected fluid material is composed of virgin PET, recycled PET or a combination of virgin PET and recycled PET, and preferably more than 90% recycled PET.

The process for manufacturing the rod carrier then comprises a step of solidifying the fluid material in the molding cavity **92** to form the rod carrier **40**.

In this aim, the fluid material is cooled to reach a temperature less than 70° C., and more particularly less than 62° C. in its core.

For this purpose, a coolant circulates in the cooling circuits **106** defined in the die **82**, the carriages **88** and the steady **90** so as to cool the molding cavity **92**.

The coolant is for example cold water, particularly water at a temperature less than 40° C., particularly less than 15° C.

Furthermore, according to the invention, during the solidification step, the central spindle **84** is cooled by injecting a coolant into the channel **102** defined therein.

In order to increase the cooling, the coolant is advantageously injected continuously, for example in fountain form, such that the coolant present in the channel **102** always remains at a low temperature.

The threaded spindle **86** is cooled by thermal contact with the central spindle **84**.

The upper ends **98**, **104** of the two spindles **84**, **86** are thus cooled and cool by thermal contact the fluid material present in the molding cavity **92** so as to solidify it.

This leads to precise manufacturing and rapid solidification of the cylindrical skirt **44**, and in particular of the upper and lower parts.

The rod carrier **40** thus formed in the molding cavity **92** is then released from the molding device **80**.

For this purpose, the steady **90** is translated along the central axis D-D' downward. The carriages **88** are then translated along a perpendicular axis to the central axis D-D' in a mutually opposite direction so as to release the spindles **84**, **86**. The die **82** and the two spindles **84**, **86** are then translated along the central axis D-D', so as to release the rod carrier **40** formed by the cooled fluid material in the molding cavity **92**.

The manufacture of a rod carrier **40** according to such a process for manufacturing the rod carrier has for example a cycle time between 20 seconds and 45 seconds.

The process for manufacturing the cosmetic product application member **16** comprises a step of providing a rod **38**.

The rod **38** is for example previously molded flat by plastic injection and more particularly by injecting the same material as the rod carrier.

More particularly, a fluid material intended to solidify is injected into a molding cavity extending along a horizontal axis, so as to form on cooling the rod **38**.

As illustrated in FIG. **5**, the rod **38** is for example manufactured in a molding device **108** composed of a first shell **110** and a second shell **112** defining a molding cavity **114** therebetween.

The molding cavity **114** extends along a horizontal axis E-E'.

A fluid material intended to solidify is injected into the molding cavity **114**, for example by a nozzle (not illustrated). Once the fluid material has solidified, the rod **38** thus formed in the molding cavity **114** extends along the horizontal axis E-E' corresponding to the rod axis C-C'.

The rod **38** is then released from the molding device **108**.

For this purpose, at least one of the shells **110**, **112** is moved in a direction opposite the other shell **112**, **110** along a mold release axis F-F' perpendicular to the horizontal axis E-E'.

For example, the second shell **112** is moved upward as illustrated by the arrows in FIG. **5**.

Unlike a mold release with a movement of the shell along the rod axis C-C', such a mold release by moving the shell(s) **110**, **112** along a mold release axis F-F' perpendicular to the rod axis C-C' prevents adhesion of the material to the shell wall and therefore stretching of the rod **38** along the axis C-C' thereof.

The process for manufacturing the cosmetic product application member **16** then comprises a step of assembling the first end **70** of the rod **38** in the rod carrier **40**.

More particularly, the first end **70** of the rod **38** is snap-locked in the rod carrier **40**, and more specifically in the cylindrical skirt **44** of the rod carrier **40**.

For this purpose, during the insertion of the rod **38** into the cylindrical skirt **44** of the rod carrier **40**, the tenons **74** at the first end **70** of the rod **38** are elastically deformed so as to be housed in the retaining housing **66**. Each tenon **74** then returns to the idle position thereof and thus remains locked in the retaining housing **66**. Such shape cooperation locks the first end **70** of the rod **38** in position in the rod carrier **40**.

Such an application member is manufactured from an easy-to-recycle material that has well-established industrial recycling processes.

Moreover, such an application member can be manufactured solely from recycled materials.

Such an application member is simple and inexpensive to manufacture. Indeed, a molding of the rod carrier comprising a cooling step by injecting a coolant into the channel **102** defined in the central spindle **84** offers the advantage of reducing the cooling time. Moreover, a snap-locking fastening of the rod carrier and the rod is simple and inexpensive.

Furthermore, such a cosmetic product packaging assembly **10**, particularly when all the elements are manufactured from the same material, is advantageously easy to recycle.

The invention claimed is:

1. A cosmetic product application member comprising:

a rod carrier;

a rod defining a first end and a second end that are opposite; and

characterized in that the first end of the rod is intended to be assembled in the rod carrier, the rod and the rod carrier being manufactured from a plastic material being composed of a copolyester; wherein the rod carrier comprises a screwing ring centered on a rod carrier axis (B-B') and defining an internal thread and a cylindrical skirt centered on the rod carrier axis, extending at least partially inside the screwing ring and intended to receive the first end of the rod.

2. The cosmetic product application member according to claim **1**, wherein the plastic material is composed of virgin polyethylene terephthalate (PET), recycled PET or a combination of virgin PET and recycled PET.

3. The cosmetic product application member according to claim **1**, wherein the first end of the rod is intended to be snap-locked in the rod carrier.

4. The cosmetic product application member according to claim **1**, wherein the cylindrical skirt comprises a protuberance centered on the rod carrier axis (B-B'), protruding inside the cylindrical skirt and defining at least one retaining housing intended to receive at least one tenon arranged at the first end of the rod.

5. The cosmetic product application member according to claim **1**, wherein the cylindrical skirt comprises longitudinal ribs arranged along at least a portion of an inner surface of the cylindrical skirt, the rod comprising at least one longitudinal protrusion capable of engaging between two adjacent longitudinal ribs of the cylindrical skirt.

6. A cosmetic product packaging assembly comprising:

a receptacle intended to receive the cosmetic product and defining an opening;

a cover intended to close the opening; and

a cosmetic product application member according to claim **1**, the rod carrier being intended to be fastened in the cover.

7. The cosmetic product packaging assembly according to claim **6**, wherein the receptacle, the cover, the rod carrier and the rod are manufactured from the same material.

8. A process for manufacturing a rod carrier, the rod carrier being intended to be assembled with a rod to form a cosmetic product application member, the rod being manufactured from a plastic material being composed of a copolyester, the manufacturing process including the following steps:

providing a rod carrier molding device comprising a die and a central spindle and defining a rod carrier molding cavity; and

injecting a fluid material into the molding cavity; and

solidifying the fluid material in the molding cavity to form the rod carrier;

characterized in that the fluid material injected into the molding cavity is a copolyester, and wherein the rod carrier comprises a screwing ring centered on a rod carrier axis (B-B') and defining an internal thread and a cylindrical skirt centered on the rod carrier axis, extending at least partially inside the screwing ring and intended to receive the first end of the rod.

9. The process for manufacturing a rod carrier according to claim **8**, wherein, during the solidification step, the central spindle is cooled by injecting a coolant into a channel defined inside the central spindle.

10. The process for manufacturing a rod carrier according to claim **8**, wherein the molding device further comprises a threaded spindle; during the solidification step, the threaded spindle being cooled by thermal contact with the central spindle.

11. The process for manufacturing a rod carrier according to claim **8**, wherein the injected fluid material is composed of virgin polyethylene terephthalate (PET), recycled PET or a combination of virgin PET and recycled PET.

12. A process for manufacturing a cosmetic product application member including the following steps:

manufacturing a rod carrier according to the manufacturing process according to claim **8**;

providing a rod defining a first end and a second end that are opposite, the rod being composed of the same material as the rod carrier; and

assembling the first end of the rod in the rod carrier.

13. The process for manufacturing a cosmetic product application member according to claim **12**, wherein the first end of the rod is snap-locked in the rod carrier.

14. The process for manufacturing a cosmetic product application member according to claim **12**, wherein the rod is previously molded flat.

15. The cosmetic product application member according to claim **2**, wherein the first end of the rod is intended to be snap-locked in the rod carrier.

16. A cosmetic product packaging assembly comprising:

a receptacle intended to receive the cosmetic product and defining an opening;

a cover intended to close the opening; and

a cosmetic product application member according to claim **2**, the rod carrier being intended to be fastened in the cover.

17. A cosmetic product packaging assembly comprising:

a receptacle intended to receive the cosmetic product and defining an opening;

a cover intended to close the opening; and

a cosmetic product application member according to claim **3**, the rod carrier being intended to be fastened in the cover.

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